

Q1 JEE Main 2020 - 7 Jan (Evening)

Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true?

- (A) $P_5 = P_2 \times P_3$
- (B) $P_1 + P_2 + P_3 + P_4 + P_5 = 25$
- (C) $P_3 = P_5 - P_4$
- (D) $P_5 = 11$

Q2 JEE Main 2020 - 8 Jan (Morning)

The least positive value of a for which the equation, $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

Q3 JEE Main 2020 - 9 Jan (Morning)

Find number of real roots of equation $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q4 JEE Main 2020 - 9 Jan (Evening)

Let $a, b \in R$, $a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to:

- (A) 25
- (B) 26
- (C) 24
- (D) 28

Q5 JEE Main 2020 - 2 Sep (Morning)

Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3, \dots$, then :

- (A) $5S_6 + 6S_5 = 2S_4$

(B) $6S_6 + 5S_5 + 2S_4 = 0$

(C) $6S_6 + 5S_5 = 2S_4$

(D) $5S_6 + 6S_5 + 2S_4 = 0$

Q6 JEE Main 2020 - 2 Sep (Evening)

Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in :

(A) $(-1, 0)$

(B) $(-3, -1)$

(C) $(0, 1)$

(D) $(1, 3)$

Q7 JEE Main 2020 - 3 Sep (Morning)

If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right) \left(\beta - \frac{1}{\beta}\right) \left(\alpha + \frac{1}{\beta}\right) \left(\beta + \frac{1}{\alpha}\right)$ is equal to :

(A) $\frac{9}{4} (9 - q^2)$

(B) $\frac{9}{4} (9 + p^2)$

(C) $\frac{9}{4} (9 + q^2)$

(D) $\frac{9}{4} (9 - p^2)$

Q8 JEE Main 2020 - 3 Sep (Morning)

Consider the two sets:

$A = \{m \in R : \text{both the roots of } x^2 - (m+1)x + m+4 = 0 \text{ are real}\}$ and $B = [-3, 5]$

Which of the following is not true?

(A) $A \cap B = \{-3\}$

(B) $B - A = (-3, 5)$

(C) $A \cup B = R$

(D) $A - B = (-\infty, -3) \cup (5, \infty)$

Q9 JEE Main 2020 - 3 Sep (Evening)

The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0,1)$ is :

- (A) $(-3,-1)$
- (B) $(2,4]$
- (C) $(0,2)$
- (D) $(1,3]$

Q10 JEE Main 2020 - 4 Sep (Morning)

Let α and β be the roots of $x^2 - 3x + p = 0$ and γ and δ be the roots of $x^2 - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ form a geometric progression. Then ratio $(2q + p) : (2q - p)$ is :

- (A) $3 : 1$
- (B) $5 : 3$
- (C) $9 : 7$
- (D) $33 : 31$

Q11 JEE Main 2020 - 4 Sep (Evening)

Let $\lambda \neq 0$ be in $\mathbb{R}$. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to :

- (A) 18
- (B) 9
- (C) 27
- (D) 36

Q12 JEE Main 2020 - 5 Sep (Morning)

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is :

- (A) $\frac{25}{9}$
- (B) $\frac{25}{81}$
- (C) $\frac{5}{9}$
- (D) $\frac{5}{27}$

Q13 JEE Main 2020 - 5 Sep (Evening)

If α and β are the roots of the equation, $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to :

- (A) $\frac{1}{24}$
- (B) $\frac{27}{32}$
- (C) $\frac{3}{8}$
- (D) $\frac{27}{16}$

Q14 JEE Main 2020 - 6 Sep (Morning)

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of

$$\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8} \text{ is}$$

- (A) 3
- (B) 2
- (C) 4
- (D) 1

Q15 JEE Main 2020 - 6 Sep (Evening)

If α and β are the roots of the equation $2 \times (2x + 1) = 1$, then β is equal to :

- (A) $2\alpha^2$
- (B) $-2\alpha(\alpha + 1)$
- (C) $2\alpha(\alpha - 1)$
- (D) $2\alpha(\alpha + 1)$

Answer Key

Q1 (A)

Q2 (8)

Q3 (A)

Q4 (A)

Q5 (A)

Q6 (A)

Q7 (D)

Q8 (D)

Q9 (D)

Q10 (C)

Q11 (A)

Q12 (B)

Q13 (D)

Q14 (B)

Q15 (B)

Q1

$$\alpha^5 = 5\alpha + 3$$

$$\beta^5 = 5\beta + 3$$

$$P_5 = 5(\alpha + \beta) + 6$$

$$= 5(1) + 6$$

$$P_5 = 11 \text{ and } P_5 = \alpha^2 + \beta^2 = \alpha + 1 + \beta + 1$$

$$P_2 = 3 \text{ and } P_3 = \alpha^3 + \beta^3 = 2\alpha + 1 + 2\beta + 1 = 2(1) + 2 = 4$$

$$P_2 \times P_3 = 12 \text{ and } P_5 = 11 \Rightarrow P_5 \neq P_2 \times P_3$$

Q2

$$D \geq 0$$

$$(a - 10)^2 - 4(2) \left(\frac{33}{2} - 2a \right) \geq 0$$

$$(a - 10)^2 - 4(33 - 4a) \geq 0$$

$$a^2 - 4a - 32 \geq 0 \Rightarrow a \in (-\infty, -4] \cup [8, \infty)$$

Q3

$$\text{Let } e^x = t \in (0, \infty)$$

Given equations

$$t^4 + t^3 - 4t^2 + t + 1 = 0$$

$$t^2 + t - 4 + \frac{1}{t} + \frac{1}{t^2} = 0$$

$$\left(t^2 + \frac{1}{t^2} \right) + \left(t + \frac{1}{t} \right) - 4 = 0$$

$$\text{Let } t + \frac{1}{t} = \alpha$$

$$(\alpha^2 - 2) + \alpha - 4 = 0$$

$$\alpha^2 + \alpha - 6 = 0$$

$$\alpha = -3, 2 \Rightarrow \alpha = 2 \Rightarrow e^x + e^{-x} = 2$$

$$x = 0 \text{ only solution}$$

Q4

$$2\alpha = \frac{2b}{a} \Rightarrow \alpha = \frac{b}{a} \text{ and } \alpha^2 = \frac{5}{a} \Rightarrow \frac{b^2}{a^2} = \frac{5}{a}$$

$$\Rightarrow b^2 = 5a \dots (i) \quad (a \neq 0)$$

$$\alpha + \beta = 2b \dots (ii)$$

$$\text{and } \alpha\beta = -10 \dots (iii)$$

$$\alpha = \frac{b}{a} \text{ is also root of } x^2 - 2bx - 10 = 0$$

$$\alpha = \frac{b}{a}, x^2 - 2bx - 10 = 0$$

$$\Rightarrow b^2 - 2ab^2 - 10a^2 = 0$$

$$\text{by (i)} \Rightarrow 5a - 10a^2 - 10a^2 = 0$$

$$\Rightarrow a = \frac{1}{4} \text{ and } b^2 = \frac{5}{4}$$

$$\alpha^2 = 20 \text{ and } \beta^2 = 5$$

$$\text{Now } \alpha^2 + \beta^2$$

$$= 5 + 20$$

$$= 25$$

Q5

$\therefore \alpha$ is a root of given equation, then

$$5\alpha^2 + 6\alpha = 2$$

$$\Rightarrow 5\alpha^6 + 6\alpha^5 = 2\alpha^4 \dots (1)$$

$$\text{Similarly } 5\beta^6 + 6\beta^5 = 2\beta^4 \dots (2)$$

Adding (1) and (2), we get

$$5S_6 + 6S_5 = 2S_4$$

Q6

$$\text{Let } f(x) = ax^2 + bx + c$$

Let roots are 3 and α

$$\text{and } f(-1) + f(2) = 0$$

$$4a + 2b + c + a - b + c = 0$$

$$5a + b + 2c = 0$$

$$f(3) = 0 \Rightarrow 9a + 3b + c = 0$$

From equation (i) and (ii)

$$\frac{a}{1-6} = \frac{b}{18-5} = \frac{c}{15-9} \Rightarrow \frac{a}{-5} = \frac{b}{13} = \frac{c}{6}$$

$$f(x) = k(-5x^2 + 13x + 6)$$

$$= -k(5x + 2)(x - 3)$$

Roots are 3 and $-\frac{2}{5}$

$-\frac{2}{5}$ lies in interval $(-1, 0)$

Q7

$$\alpha \cdot \beta = 2 \text{ and } \alpha + \beta = -p \text{ also } \frac{1}{\alpha} + \frac{1}{\beta} = -q$$

$$\Rightarrow p = 2q$$

$$\text{Now } \left(\alpha - \frac{1}{\alpha}\right) \left(\beta - \frac{1}{\beta}\right) \left(\alpha + \frac{1}{\beta}\right) \left(\beta + \frac{1}{\alpha}\right)$$

$$= \left[\alpha\beta + \frac{1}{\alpha\beta} - \frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right] \left[\alpha\beta + \frac{1}{\alpha\beta} + 1 + 1\right]$$

$$= \frac{9}{2} \left[\frac{5}{2} - \frac{\alpha^2 + \beta^2}{2}\right] = \frac{9}{4} [5 - (p^2 - 4)]$$

$$= \frac{9}{4} (9 - p^2)$$

Q8

$A = \{m \in R : x^2 - (m+1)x + m + 4 = 0 \text{ has real roots}$

$$A = \{(-\infty, -3] \cup [5, \infty)\} \{D \geq 0\}$$

$$B = [-3, 5) \Rightarrow A - B = (-\infty, -3) \cup [5, \infty)$$

Q9

$$\therefore \text{Equation is : } (\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$$

$\therefore$ One root in interval $(0, 1)$

$$\therefore f(0) \cdot f(1) < 0$$

$$2 \cdot (\lambda^2 + 1 - 4\lambda + 2) < 0$$

$$(\lambda - 3)(\lambda - 1) < 0$$

$$\therefore \lambda \in (1, 3)$$

If $\lambda = 3$, then roots are 1 and $\frac{1}{5}$

$$\therefore \lambda \in (1, 3]$$

Q10

$\because \alpha, \beta, \gamma, \delta$ are in G.P, so $\alpha\delta = \beta\gamma$

$$\Rightarrow \frac{\alpha}{\beta} = \frac{\gamma}{\delta} \Rightarrow \left| \frac{\alpha-\beta}{\alpha+\beta} \right| = \left| \frac{\gamma-\delta}{\gamma+\delta} \right|$$

$$\Rightarrow \frac{\sqrt{9-4p}}{3} = \frac{\sqrt{36-4q}}{6}$$

$$\Rightarrow 36 - 16p = 36 - 4q$$

$$\Rightarrow q = 4p$$

$$\text{So } \frac{2q+p}{2q-p} = \frac{9p}{7p} = \frac{9}{7}$$

Q11

Roots of $x^2 - x + 2\lambda = 0$ are α and β
and roots of $3x^2 - 10x + 27\lambda = 0$ are α and γ

Here,

$$3\alpha^2 - 10\alpha + 27\lambda = 0$$

$$3\alpha^2 - 3\alpha + 6\lambda = 0$$

$$\therefore \alpha = 3\lambda$$

Now,

$$3\lambda + \beta = 1 \text{ and } 3\lambda \cdot \beta = 2\lambda$$

$$\text{and, } 3\lambda + \gamma = \frac{10}{3} \text{ and } 3\lambda \cdot \gamma = 9\lambda$$

$$\therefore \gamma = 3, \alpha = \frac{1}{3} \text{ and } \beta = \frac{2}{3}, \lambda = \frac{1}{9}$$

$$\frac{\beta\gamma}{\lambda} = 18$$

Q12

Let $|x| = t$ we have

$$9t^2 - 18t + 5 = 0$$

$$9t^2 - 15t - 3t + 5 = 0$$

$$(3t - 1)(3t - 5) = 0$$

$$\Rightarrow t = \frac{1}{3} \text{ or } \frac{5}{3} \Rightarrow |x| = \frac{1}{3} \text{ or } \frac{5}{3}$$

$$\text{Roots are } \pm \frac{1}{3} \text{ and } \pm \frac{5}{3}$$

$$\text{Product} = \frac{25}{81}$$

Q13

$$7x^2 - 3x - 2 = 0 \Rightarrow \alpha + \beta = \frac{3}{7}, \alpha\beta = \frac{-2}{7}$$

$$\begin{aligned} \text{Now } \frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2} &= \frac{\alpha - \alpha\beta(\alpha + \beta) + \beta}{1 - (\alpha^2 + \beta^2) + (\alpha\beta)^2} = \frac{(\alpha + \beta) - \alpha\beta(\alpha + \beta)}{1 - (\alpha + \beta)^2 + 2\alpha\beta + (\alpha\beta)^2} \\ &= \frac{\frac{3}{7} + \frac{2}{7} \times \frac{3}{7}}{1 - \frac{9}{49} + 2 \times \frac{-2}{7} + \frac{4}{49}} = \frac{21+6}{49-9-28+4} = \frac{27}{16} \end{aligned}$$

Q14

$$\frac{\alpha^{3/8}}{\beta^{5/8}} + \frac{\beta^{3/8}}{\alpha^{5/8}} = \frac{\alpha + \beta}{(\alpha\beta)^{5/8}}$$

$$\text{For } x^2 - 64x + 256 = 0$$

$$\alpha + \beta = 64$$

$$\alpha\beta = 256$$

$$\therefore \frac{\alpha + \beta}{(\alpha\beta)^{5/8}} = \frac{64}{(2^8)^{5/8}} = \frac{64}{32} = 2$$

Q15

$$\alpha + \beta = -\frac{1}{2} \Rightarrow -1 = 2\alpha + 2\beta$$

$$\text{and } 4\alpha^2 + 2\alpha - 1 = 0$$

$$\Rightarrow 4\alpha^2 + 2\alpha + 2\alpha + 2\beta = 0$$

$$\Rightarrow \beta = -2\alpha(\alpha + 1)$$